

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian	Molecules Electroporated	DNA: plasmid.
Species Used	Human, V79, skin cells, fibroblasts		

Before the Pulse

Cell Growth Medium	Modified MEM, 5%Fetal Calf Serum (GIBCO/BRL, Sigma)	Growth Phase at Harvest	Actively growing 70% confluent
		Pre-pulse Incubation	10 min / ice
Wash Solution	Phosphate Buffered Saline		

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature	Room temperature	Cuvette Gap	0.4 cm
Electroporation Medium*	10 mM HEPES	Voltage	0.450 kV
Cell Density	10 (7) cells / ml	Field Strength	1.125 kV/cm
Volume of Cells	0.4 ml	Capacitor	25 µF
DNA Concentration	20 µg	Resistor	5 Ω
DNA Resuspension Buffer	Not given	Time Constant	10 msec
Volume of DNA	10 µl		
After the Pulse			
Outgrowth Medium	Not given		

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	37 °C
Length of Incubation	1 week
Selection Method or Assay Used	G418 or ampicillin
Electroporation Efficiency	10 (6) transformants / µg
Per Cent Survival	2%

Name of Submitter Phillip K. Liu, Ph.D., Assistant Professor

Institution Address M.D. Anderson Cancer Center
Laboratory Medicine Box 72
1515 Halcombe Blvd.
Houston, Texas 77030

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